RVCalPy radial-velocity benchmark report

Broadening function: LL Aqr and AK For · Cross-correlation: 16 UVES high-velocity stars
RVCalPy-26: Bayesian broadening-function velocities and errors · https://github.com/atemelci/RVCalPy · 2026 November 3

We compare RVCalPy velocities with published velocities of the same observations (LL Aqr, AK For) and of the same spectra (UVES sample). In RVCalPy-26 the broadening-function (BF) velocity and its error come from the posterior of the BF profile, fitted to each spectrum (Section 2); the BF window is 500–550 nm, as before. With the BF, the LL Aqr differences scatter by 0.017 km s^{-1} for the primary and 0.033 km s^{-1} for the secondary (RVCalPy-24: 0.039 and 0.055), and the AK For differences by 0.127 and 0.134 km s^{-1} (RVCalPy-24: 0.261 and 0.390). An orbit fitted to our own velocities leaves residuals of 0.025/0.044 km s^{-1} for LL Aqr and 0.095/0.140 km s^{-1} for AK For, 13–61% less than with RVCalPy-24. Its reduced χ^2 is 1.85 and 3.39: the scatter that the per-epoch errors do not explain is a shift that both stars of a spectrum share, which no single spectrum can show; the RVCalPy-26 orbit step carries it with a model error sampled with the orbit. With cross-correlation, the 16 UVES velocities differ from the published ones by a median |Δ| of 0.483 km s^{-1} and a weighted offset of +0.181 ± 0.126 km s^{-1}. Over the 919 km s^{-1} velocity span the slope is 1.000183 ± 0.000484.

1. What was tested

	LL Aqr	AK For	UVES sample
Method	Broadening function	Broadening function	Cross-correlation
Program	RVCalPy-26	RVCalPy-26	RVCalPy-26
RV error	Bayesian: posterior median, 16th–84th percentiles	Bayesian: posterior median, 16th–84th percentiles	larger of width error and chunk error
Reference	Graczyk et al. (2016)	Hełminiak et al. (2014)	Alazzawi et al. (2026)
Instrument	HARPS	FEROS (14) + HARPS (2)	UVES, DIC1 390+580, red arm
Measured / compared	17 / 16 observations	16 / 14 observations	16 / 16 stars, one epoch each
Template	HIP 57757 (6093 K)	HIP 108870 (4676 K)	synthetic, 5215 K, log g 3.55
Window / step	500–550 nm, Δv = 0.65 km s^{-1} (FWHM/4)	500–550 nm, Δv = 1.71 (FEROS), 0.65 (HARPS) km s^{-1}	500–550 nm
Period / velocity span	P = 20.1784 d	P = 3.98099 d	919 km s^{-1}

Table 1. Material and settings. All spectra are public ESO archive phase-3 products; only epochs present in both data sets are compared. The settings are those of the RVCalPy-24 report; the broadening-function velocities and errors are the Bayesian ones of RVCalPy-26, which also leaves zero-filled detector gaps out of the SVD and sets the component labels again from the posterior.

2. How the numbers are defined

For each matched epoch we take the difference between our velocity and the published one, and add the two uncertainties in quadrature. Δ is the weighted mean offset and s the scatter. The slope a and intercept b come from a weighted straight-line fit, and a − 1 is quoted as the amplitude error (BF) or the velocity-scale error (CCF):

$$d_i = v_i^{\mathrm{ours}} - v_i^{\mathrm{pub}}\,, \quad \sigma_i = \sqrt{\varepsilon_i^{2,\,\mathrm{ours}} + \varepsilon_i^{2,\,\mathrm{pub}}}\,, \quad w_i = \sigma_i^{-2}$$

$$\Delta = \frac{\sum w_i d_i}{\sum w_i}\,, \quad s = \sqrt{\frac{\sum (d_i - \bar{d})^2}{N-1}}\,, \quad v^{\mathrm{ours}} = a\, v^{\mathrm{pub}} + b$$

Broadening-function velocities and errors (since RVCalPy-25). The BF is the kernel of a linear model: the line depths y of the spectrum are the template line depths, shifted to every velocity u_j of the BF grid (the design matrix D) and weighted by the BF (Rucinski 1992, 2002). RVCalPy fits the profile of each star, a Gaussian kernel K of area a_k, velocity v_k and width w_k, through this model to the spectrum itself, not to the smoothed BF, with a straight-line continuum $c_0 + c_1 x$ in each 10 nm segment s of the window:

$$y_t = \sum_j D_{tj}\, b_j(\theta) + c_0 + c_1 x_t + e_t\,, \quad b_j(\theta) = \sum_k a_k\, K(u_j;\, v_k, w_k)$$

This is the BF with its full noise covariance, so neither the BF smoothing nor the SVD cut enters the result. The noise e is correlated, because template mismatch leaves residuals that are correlated over a line width: each segment has its own autoregressive model AR(p_s), order chosen by BIC (Berk 1974), as Czekala et al. (2015) recommend for imperfect spectral models. The continuum (flat prior) and the noise level of each

segment (Jeffreys prior) are integrated out analytically, which leaves the posterior

$$\ln p(\theta \mid y) = -\sum_s \frac{n_s - 2}{2} \ln Q_s(\theta) + \ln p(\theta)\,, \quad e_t \sim \mathrm{AR}(p_s)$$

with Q_s the whitened, continuum-free sum of squared residuals of segment s (cf. Asensio Ramos & Petit 2015 for the Bayesian form of the least-squares deconvolution, and Silva et al. 2022 for a per-spectrum RV posterior). emcee (Foreman-Mackey et al. 2013) samples it; **the velocity is the posterior median and its error ε half the 16th–84th percentile range**. Zero-flux detector gaps (HARPS 530.4–533.7 nm) are left out, and the kernel is never narrower than half a BF bin, which keeps the velocities from locking onto the grid. On synthetic SB2 spectra of known velocity (RVCalPy-25 code, the same posterior; S/N 80, this window, the LL Aqr template) these errors are calibrated: 35 epochs gave a scatter of 4.9/5.8 m s⁻¹ for errors of 4.4/6.6 m s⁻¹ (pull rms 1.10/0.87; 0.95/0.63 with a deliberately mismatched template), while the profile fit of the smoothed BF used up to RVCalPy-24 scattered 75/111 m s⁻¹ for errors of 37/55 m s⁻¹ (pull rms 1.91/1.91). With RVCalPy-26, synthetic copies of three well-separated LL Aqr epochs, built from the fitted model with white noise at the photon level of each spectrum, give a pull rms of 1.05 ± 0.09 over 60 velocities, with 67% of the pulls within ±1 and 93% within ±2.

The differences to the published velocities do not test our error bars: both analyses of a star measured the same spectra, so the noise of each spectrum is in both velocities and largely cancels in their difference. For LL Aqr the reduced chi-square of the differences would be 0.09 and 0.25 even if each analysis' errors were exactly its own orbit-fit rms. Our errors ε are therefore tested against an orbit fitted to our velocities alone (RVCalPy orbit code, least squares; LL Aqr eccentric with one systemic velocity per star, as in Graczyk et al. 2016; AK For circular with a zero-point offset for HARPS). χ^2_r is its reduced chi-square over both components and f the factor by which the errors fall short; the RVCalPy-26 orbit step covers this scatter with a model error sampled with the orbit, and applies f itself only with --model-error scale:

$$r_i = v_i^{\mathrm{ours}} - v^{\mathrm{orbit}}(t_i)\,, \quad \chi^2_r = \frac{1}{2N - n_p} \sum_{i,\ k=1,2} \left(\frac{r_i}{\varepsilon_i^{\mathrm{ours}}}\right)^2, \quad f = \sqrt{\chi^2_r}$$

No χ^2_r is given for the UVES comparison: both velocities are measured from the same exposures and no per-star error is published (the paper quotes a global ~0.5 km s⁻¹).

3. Broadening function: LL Aqr and AK For

	LL Aqr		AK For	
	Primary	**Secondary**	**Primary**	**Secondary**
Observations compared	16	16	14	14
Offset Δ (km s⁻¹)	+0.287 ± 0.010	+0.274 ± 0.019	+0.193 ± 0.021	+0.261 ± 0.057
Scatter s (km s⁻¹)	0.017	0.033	0.127	0.134
RVCalPy-24	0.039	0.055	0.261	0.390
Slope a	1.0000	1.0001	1.0015	0.9993
Amplitude error	+0.00%	+0.01%	+0.15%	-0.07%
Median error ε (km s⁻¹)	0.020	0.038	0.050	0.104
Own orbit: rms (km s⁻¹)	0.025	0.044	0.095	0.140
RVCalPy-24	0.039	0.051	0.189	0.360
Own orbit: χ^2r, f	1.85, 1.36		3.39, 1.84	
RVCalPy-24	0.83, 0.91		2.91, 1.71	
Own orbit: corr(r_1, r_2)	+0.82 (p < 0.001)		+0.43 (p = 0.129)	
Own orbit: K (km s⁻¹)	49.925 ± 0.009	57.750 ± 0.016	70.554 ± 0.038	76.990 ± 0.069
Authors' K (km s⁻¹)	49.948 ± 0.013	57.736 ± 0.014	70.471 ± 0.031	77.136 ± 0.051
Authors' RVs, our model: K	49.928 ± 0.009	57.752 ± 0.017	70.436 ± 0.022	77.098 ± 0.063
Authors' own fit rms	0.053	0.049	0.082	0.096

Table 2. Broadening-function velocities against the published ones (upper part) and against an orbit fitted to our own velocities (lower part; χ^2_r, f and corr(r_1, r_2), the correlation of the two stars' residuals over the epochs, belong to the whole orbit; K is given with the least-squares errors after the error scale f). Measured with RVCalPy-26; the grey rows give the same quantity for RVCalPy-24 (profile fit of the smoothed BF) on the same spectra with the same settings. The last rows give the authors' own K, the K of the authors' velocities of the same epochs fitted with our orbit model, and the authors' orbit-fit rms.

LL Aqr (16 observations). For the primary and secondary respectively, the offsets are +0.287 ± 0.010 and +0.274 ± 0.019 km s⁻¹, the scatter is 0.017 and 0.033 km s⁻¹, and the amplitude errors are +0.00% and +0.01%. The offsets changed by +0.09 and +0.05 km s⁻¹ from RVCalPy-24: a zero point that is the same at

every epoch, which an orbit takes into γ. Our own orbit leaves residuals of 0.025 and 0.044 km s⁻¹ (RVCalPy-24: 0.039 and 0.051) with χ^2_r = 1.85. The residuals of the two stars move together (correlation +0.82): one spectrum shifts both stars by about 0.014–0.023 km s⁻¹ beyond the errors, which a per-epoch error cannot contain; the model error of the RVCalPy-26 orbit step carries it (f = 1.36). $\gamma_2 - \gamma_1$ is +0.257 ± 0.018 km s⁻¹; Graczyk et al. give +0.195 ± 0.012 from their full solution, and their velocities fitted with our model give +0.250 ± 0.018.

AK For (14 observations: 12 FEROS, 2 HARPS). For the primary and secondary respectively, the offsets are +0.193 ± 0.021 and +0.261 ± 0.057 km s⁻¹, the scatter is 0.127 and 0.134 km s⁻¹, and the amplitude errors are +0.15% and -0.07%. Our own orbit leaves residuals of 0.095 and 0.140 km s⁻¹ (RVCalPy-24: 0.189 and 0.360) with χ^2_r = 3.39 (with the two FEROS epochs that the published table does not list, χ^2_r = 6.13). The scatter beyond the errors, about 0.08–0.09 km s⁻¹, is similar for both stars (correlation +0.43) and close to the night-to-night zero-point drift of a spectrograph without simultaneous calibration such as FEROS; the orbit model error carries it (f = 1.84). The secondary differences run from +0.10 km s⁻¹ at phase 0.37 to +0.56 km s⁻¹ at phase 0.90.

K. With both errors combined, our K differ from the authors' by -1.5σ, +0.6σ, +1.7σ, -1.7σ (LL Aqr primary, secondary; AK For primary, secondary). Fitted with our orbit model on the same epochs, the authors' own velocities give K = 49.928/57.752 km s⁻¹ for LL Aqr and 70.436/77.098 km s⁻¹ for AK For. Ours differ from these by -0.003/-0.002 and +0.118/-0.109 km s⁻¹: for LL Aqr the two velocity sets give the same orbit, and the difference to the published K comes from the orbit solutions; for AK For it is the amplitude difference of the two velocity sets (slopes 1.0015 and 0.9993), a template and method systematic of the same spectra rather than noise.

4. Cross-correlation: 16 UVES high-velocity stars

Star	RV, this work [km s⁻¹]	RV, previous work [km s⁻¹]	RV, Gaia DR3 [km s⁻¹]	Δ [km s⁻¹]
TYC 7524–00065-1	+351.487 ± 0.047	+351.920	+350.953 ± 0.217	-0.433
TYC 7535–00160-1	+355.061 ± 0.075	+353.523	+355.069 ± 0.253	+1.538
C0213360-505024	+387.973 ± 0.063	+388.600	+388.222 ± 0.611	-0.627
C1012254-203007	+508.080 ± 0.085	+506.755	+508.138 ± 1.068	+1.325
C1032508-244851	+509.635 ± 0.096	+509.490	+515.354 ± 2.470	+0.145
C1126364-293415	+411.112 ± 0.023	+409.870	+409.674 ± 0.926	+1.242
C1250398-030748	+398.196 ± 0.100	+403.020	+396.221 ± 0.979	-4.824
C1302091-323721	+410.533 ± 0.035	+410.000	+409.817 ± 3.027	+0.533
C1304064-331913	+360.415 ± 0.034	+360.720	+357.825 ± 0.994	-0.305
C1330590-230955	+335.503 ± 0.066	+335.350	+334.919 ± 0.526	+0.153
TYC 7274–00734-1	+408.415 ± 0.048	+408.317	+407.955 ± 0.208	+0.098
C1519196-191359	-409.450 ± 0.082	-409.610	-409.694 ± 6.570	+0.160
C1536201-144228	-329.661 ± 0.079	-329.600	-328.647 ± 2.112	-0.061
C1919566-632839	+330.789 ± 0.048	+328.680	+330.399 ± 0.530	+2.109
C2136142-694908	+383.363 ± 0.024	+381.640	+382.882 ± 0.330	+1.723
TYC 8019–00159-1	+99.092 ± 0.098	+99.117	+98.450 ± 0.171	-0.025

Table 3. Rows follow the order of Table 3 in Alazzawi et al. (2026). Our velocities are barycentric, measured with RVCalPy-26 on the cleaned spectrum; ± is the larger of the CCF width error (Zucker 2003) and the chunk error of five wavelength parts. RVCalPy-25 and RVCalPy-26 change only the broadening function and the orbit step: these CCF velocities and errors are identical to those of RVCalPy-24. “RV, previous work” is the template-matching velocity of Alazzawi et al. (2026) from the same spectrum, for which no per-star error is published, and Δ is this work minus that velocity. The Gaia DR3 velocity is listed for reference only and enters no statistic.

	All 16 stars
Weighted offset Δ (km s⁻¹)	+0.181 ± 0.126
Median |Δ| (km s⁻¹)	0.483
Scatter s (km s⁻¹)	1.569
Within 1 km s⁻¹	10 of 16
Velocity range (km s⁻¹)	-409.4 to +509.6
Slope a	1.000183 ± 0.000484
Velocity-scale error a − 1	+0.018 ± 0.048%

Table 4. Summary of Table 3.

All 16 stars. The weighted offset is +0.181 ± 0.126 km s⁻¹, the median |Δ| is 0.483 km s⁻¹, and the scatter is 1.569 km s⁻¹. 8 stars agree within 0.5 km s⁻¹ and 10 within 1 km s⁻¹. Across -409.450 to +509.635 km s⁻¹ (919 km s⁻¹), the slope is 1.000183 ± 0.000484, a velocity-scale error of +0.018 ± 0.048%.

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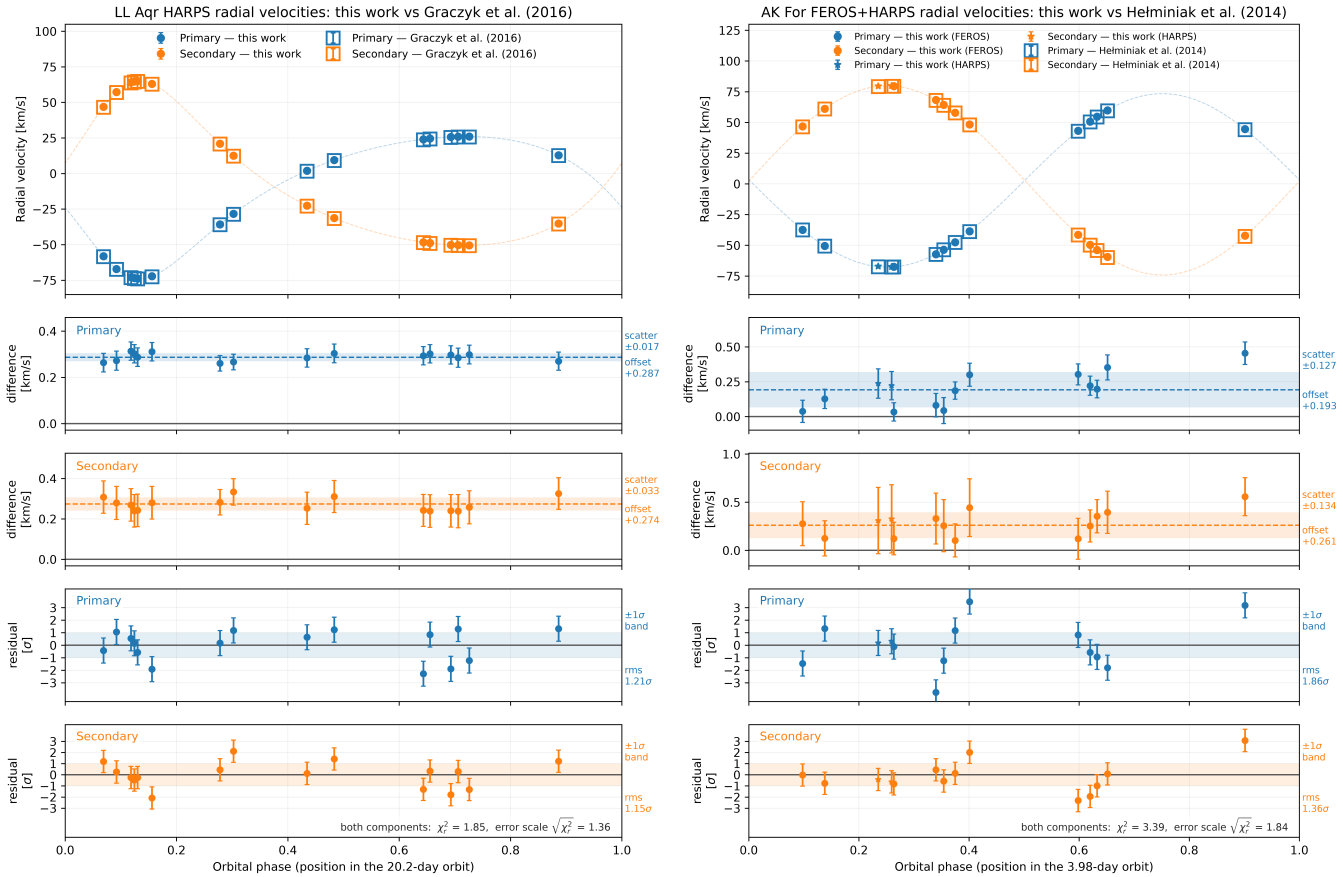


Figure 1. Broadening-function velocities of RVCalPy-26 against the published ones: LL Aqr versus Graczyk et al. 2016 (left, 16 observations) and AK For versus Hełminiak et al. 2014 (right, 14 observations; circles FEROS, stars HARPS). Top: phased velocity curves (filled: our velocities, open: published; faint dashed lines: our orbit). Middle: our velocity minus the published one, with the weighted offset (dashed) and the scatter (band), both beside each panel. Bottom: residuals of our velocities from our orbit in units of our Bayesian 1σ errors, with their rms beside each panel and the reduced χ^2 and error scale of our orbit in the last panel.